

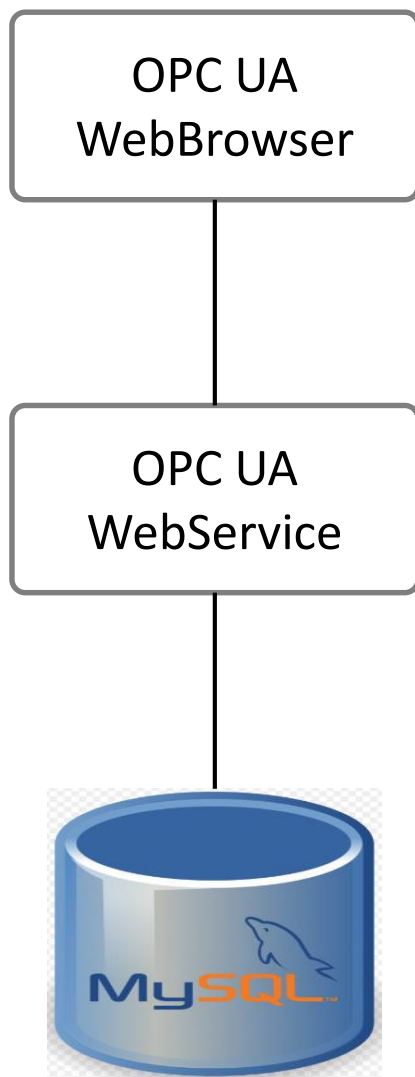


OPC UA WebAPI开发教程

— 开发环境测试

- 1.编译opcua-webservice-spring类库
- 2.编译uaweb servicetest测试程序
- 3.使用opcua-webbrowser测试服务

— 课程介绍



▼ Servers

demo

▼ Objects

Server

▼ Employees

▼ Zhang

▼ Li

PersonalData

EmployeeData

SkillInfo

▼ Wang

▼ Zhao

▼ Sun

▼ Companies

Research center

▼ Types

▼ Views

VariablesReferences

Index	Name	Data Type	Value
1	Sex	Sex	0(Male)
2	Birthday	DateTime	1993/7/12 00:00:00
3	PhoneNumber	String	XXXXXXXXXXXX
4	Address	String	XX,Shanghai

GetServiceTime

Salary

EmployeeData

EmployeeDataModel

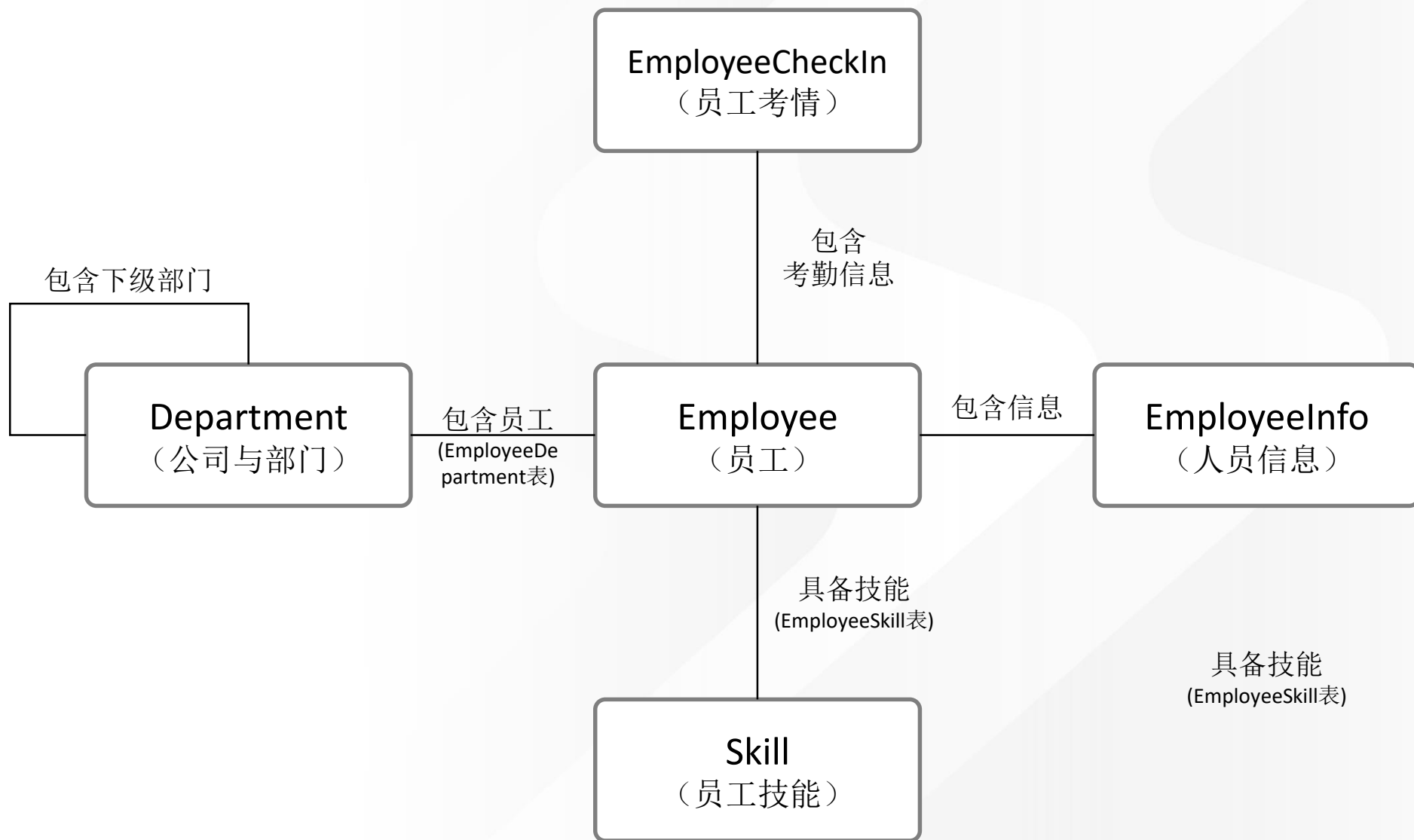
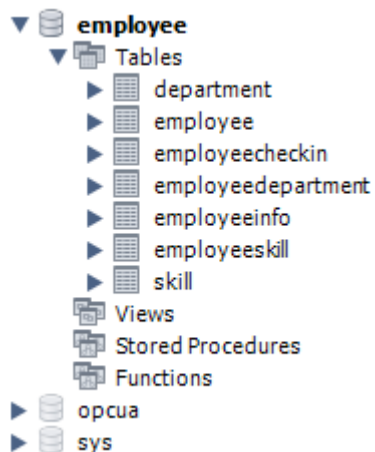
StartTime

HasComponent

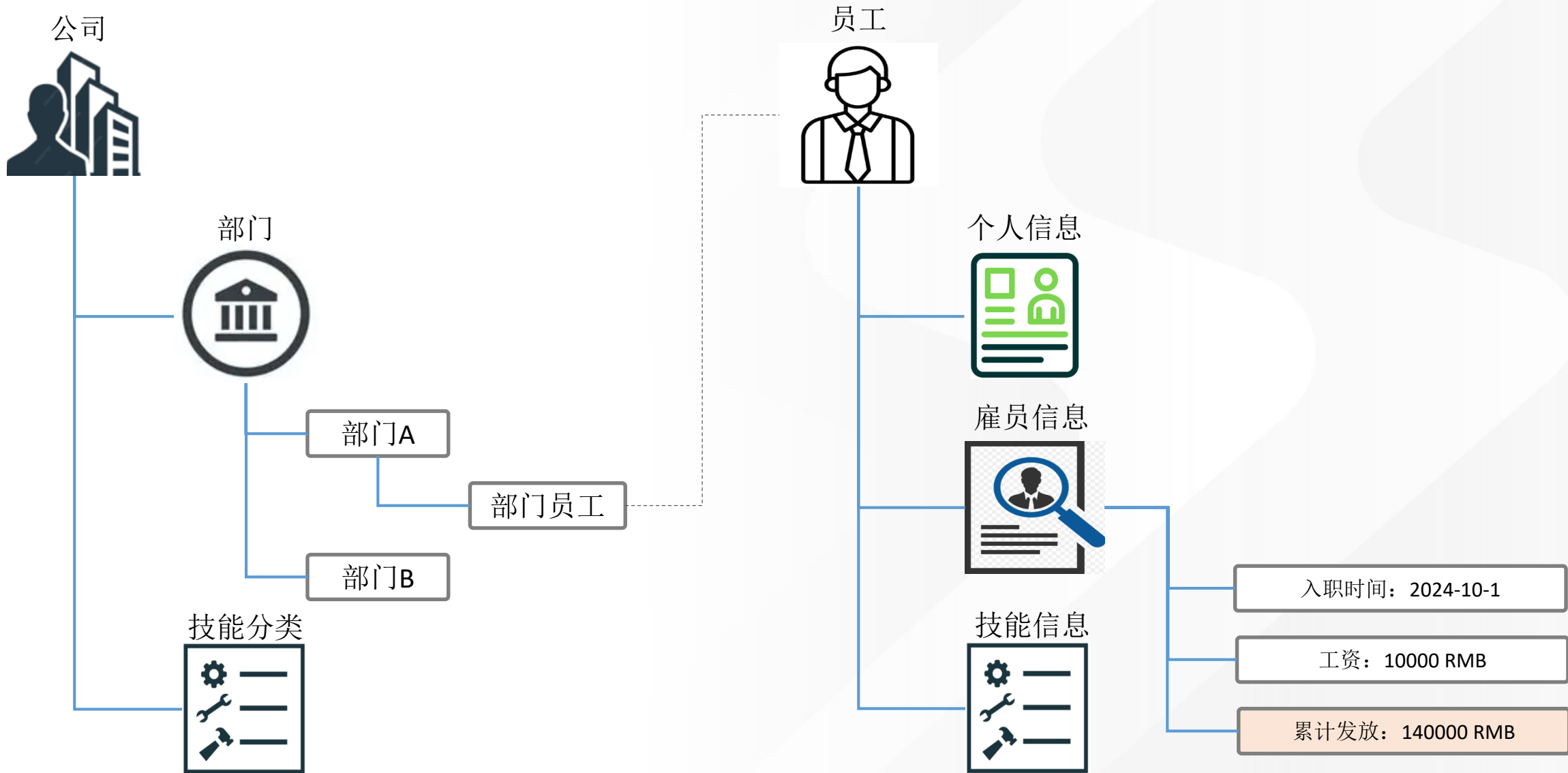
HasTypeDefinition

IsSubTypeOf

— 课程介绍-数据源

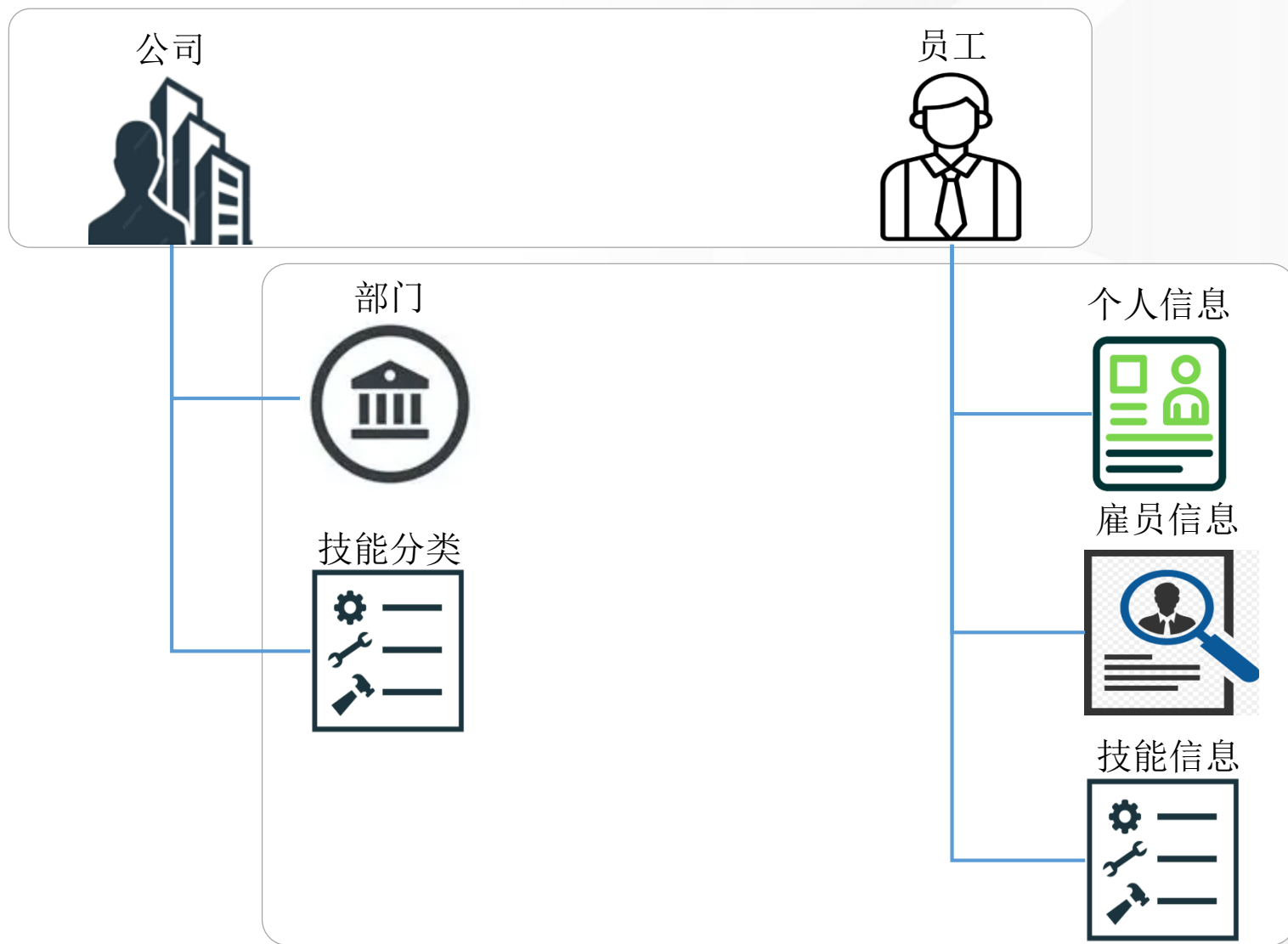


— 概念介绍



— 概念介绍

数字孪生(Digital twin)



— 概念介绍

雇员信息



元素(Element)

入职时间: 2024-10-1

工资: 10000 RMB

累计发放: 140000 RMB

数据元素(Property element): 又称变量, 用于提供字段数值

操作元素(Operation element): 又称方法, 用于提供功能调用

部门



部门A

部门员工

部门B

元素集合(Element collection): 包含子元素的对象, 且结构固定

引用元素(Reference element): 又称链接, 用于指向其它对象

— 概念介绍

数字孪生



Digital twin: 某一事物的数字呈现，并由若干子模型(Submodel)负责其各个方面的信息和能力



Digital twin repository: 用于管理一组数字孪生的目录

子模型



Submodel: 负责展现某一方面的信息或能力，由固定结构的元素构成

Dynamic Submodel: 作用同上，但元素结构不固定

元素



Property: 数值元素，对应数据库中的一个字段或时序数据库中的数据点



Operation: 操作元素，对应一段可被调用的后台代码



Reference: 引用元素，对应一组关联关系，如部门和员工



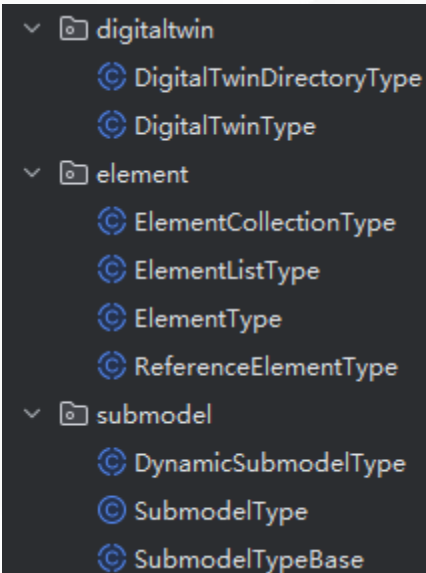
Event: 事件元素，对应一组事件，如薪水发放，上下班打卡等日志



Collection: 元素集合，由结构固定的一组子元素组成



Element list: 元素列表，由一组子元素组成，但结构不固定



— 课程1

▼ Servers

demo

▼ Objects

Server

Employees

▼ Zhang

PersonalData

▼ EmployeeData

GetCumulatedSalary

Li

Wang

Zhao

Sun

VariablesReferences			
Index	Name	Data Type	Value
1	Sex	Sex	1(Female)
2	Birthday	DateTime	1990/6/1 00:00:00
3	PhoneNumber	String	XXXXXXXXXXXX
4	Address	String	XX,Beijing

ID	Name	ID	Sex	Birthday	PhoneNumber	Address	StartTime	Salary	EmployeeId
1	Zhang	1	1	1990-06-01 00:00:00	XXXXXXXXXXXX	XX,Beijing	2020-01-01 00:00:00	10000	1
2	Li	2	0	1993-07-12 00:00:00	XXXXXXXXXXXX	XX,Shanghai	2021-01-01 00:00:00	11000	2
3	Wang	3	1	1987-04-23 00:00:00	XXXXXXXXXXXX	XX,Hangzhou	2015-09-01 00:00:00	12000	3
4	Zhao	4	0	1999-01-25 00:00:00	XXXXXXXXXXXX	XX,HaErBin	2024-09-01 00:00:00	13000	4
5	Sun	5	1	1997-12-19 00:00:00	XXXXXXXXXXXX	XX,Tianjin	2022-06-01 00:00:00	14000	5
NULL	NULL								

— 课程1

可视化

员工数字孪生定义

- ▼ DigitalTwinType
 - ▶ Employee
- ▼ DigitalTwinRepositoryType
 - ▶ EmployeeRepository

员工数字孪生

- ▼ Objects
 - ▶ Server
 - ▼ Employees
 - ▶ Zhang
 - ▶ Li
 - ▶ Wang
 - ▶ Zhao
 - ▶ Sun

EmployeeDigitalTwinType

- 用于表示员工的数字孪生
- 继承自DigitalTwinType
- 在构造函数中添加PersonalData和EmployeeData两个子模型
- 重载onGetDescriptor，并调用数据库，返回数字孪生的名称和描述信息

```
public class EmployeeDigitalTwinType extends DigitalTwinType { 4个月
    public EmployeeDigitalTwinType( 1个用法 @dingyan <dy_100000@163.
        PersonalDataSubmodelType personalDataSubmodelType,
        EmployeeDataSubmodelType employeeDataSubmodelType,
        EmployeeTwinSpace twinSpace)
    {...]}

    @Override 0个用法 @dingyan <dy_100000@163.com>
    public CompletableFuture<GetDescriptorResponse> onGetDescriptor()
}
```

EmployeeRepositoryType

- 负责提供员工数字孪生列表
- 继承自DigitalTwinRepositoryType
- 重载onGetDigitalTwinList，并调用数据库，返回员工数字孪生列表

```
// Defining entry of employee list
public class EmployeeRepositoryType extends DigitalTwinRepositoryType { 3个用法 @dingyan <dy_100000@163.com>
    public EmployeeRepositoryType(EmployeeTwinSpace twinSpace) { 1个用法 新
        super( typeld: "EmployeeRepositoryType", new LocalizedText("EmployeeRepository"), twinSpace);
    }

    @Override 0个用法 @dingyan <dy_100000@163.com>
    public CompletableFuture<GetDigitalTwinListResponse> onGetDigitalTwinList(GetDigitalTwinListReq
    {...]}
}
```

— 课程1

可视化

子模型定义

- SubmodelType
 - PersonalDataModel
 - EmployeeDataModel

个人信息

Index	Name	DataType	Value
1	Sex	Sex	1(Female)
2	Birthday	DateTime	1990/6/1 00:00:00
3	PhoneNumber	String	XXXXXXXXXXXXXX
4	Address	String	XX,Beijing

性别枚举

- Enumeration
 - Sex

累计薪资操作

- EmployeeData
 - GetCumulatedSalary

员工子模型

- Zhang
 - PersonalData
 - EmployeeData

员工信息

Index	Name	DataType	Value
1	StartTime	DateTime	2020/1/1 00:00:00
2	Salary	Float	10000
	EngineeringUnits	EUIInformation	UnitId: -1, DisplayName: ("Locale": "en", "Text": "RMB")

累计薪资执行

Call Method		
Input Arguments		
Name	Value	DataType
Name	Salary	Float
CumulatedSalary	10000	Float
Output Arguments		
Name	Value	DataType
Result	OK	

PersonalDataSubmodelType

- 用于表示员工的个人信息
- 继承自SubmodelType
- 在构造函数中添加sex, birthday等属性
- 重载onReadPropertyValues, 并调用数据库, 返回属性值
- 定义SexEnumType, 定义性别枚举类型

```
public class PersonalDataSubmodelType extends SubmodelType { 5 个用法 @dingyan
    private final UaVariable sex; 2 个用法
    private final UaVariable birthday; 2 个用法
    private final UaVariable phoneNumber; 2 个用法
    private final UaVariable address; 2 个用法

    public PersonalDataSubmodelType( 1 个用法 @dingyan <dy_100000@163.com>
        SexEnumType sexEnumType,
        EmployeeTwinSpace twinSpace) {...}

    @Override 0 个用法 @dingyan <dy_100000@163.com>
    public CompletableFuture<ReadPropertyValuesResponse> onReadPropertyValues(
        {...}
    )
}
```

EmployeeDataSubmodelType

- 用于表示员工的企业信息, 需继承SubmodelType, 并在构造函数中添加startTime, salary等
- 重载onReadPropertyValues, 并调用数据库, 返回属性值
- 重载onInvokeOperation, 调用数据库, 并计算出累计薪资

```
public class EmployeeDataSubmodelType extends SubmodelType { 5 个用法 @dingyan
    private final UaVariable startTime; 2 个用法
    private final UaVariable salary; 3 个用法

    private UaMethod getCumulatedSalary; 2 个用法

    public EmployeeDataSubmodelType(EmployeeTwinSpace twinSpace) {...}

    @Override 0 个用法 @dingyan <dy_100000@163.com>
    public CompletableFuture<ReadPropertyValuesResponse> onReadPropertyValues(
        {...}
    )

    public CompletableFuture<InvokeOperationResponse> onInvokeOperation(Invoke
    )
}
```

— 课程1

- EmployeeTwinSpace负责管理与运行员工数字孪生中的类型定义和处理逻辑
- 继承DigitalTwinSpace，并重载onStartUp，将各数字孪生定义添加到EmployeeTwinSpace中
- 通过addRepository，将DigitalTwinRepository设置到可视化界面的入口查询节点

```
public class EmployeeTwinSpace extends DigitalTwinSpace { 13 个用法  @dingyan <dy_100000@163.com> *
    // Employee definitions
    public SexEnumType sexEnumType; 2 个用法
    public EmployeeRepositoryType employeeRepositoryType; 2 个用法
    public EmployeeDigitalTwinType employeeDigitalTwinType; 2 个用法
    public PersonalDataSubmodelType personalDataSubmodelType; 2 个用法
    public EmployeeDataSubmodelType employeeDataSubmodelType; 2 个用法

    public EmployeeTwinSpace() { super( namespaceUri: "EmployeeTwinSpace"); }

    @Override 0 个用法  @dingyan <dy_100000@163.com> *
    public void onStartUp() throws UaRuntimeException {
        // Employee definitions
        sexEnumType = new SexEnumType( twinSpace: this);
        personalDataSubmodelType = new PersonalDataSubmodelType(sexEnumType, twinSpace: this);
        employeeDataSubmodelType = new EmployeeDataSubmodelType( twinSpace: this);
        employeeDigitalTwinType = new EmployeeDigitalTwinType(personalDataSubmodelType, employeeDataS
        employeeRepositoryType = new EmployeeRepositoryType( twinSpace: this);

        // Repository
        addRepository(employeeRepositoryType, id: "Employees", new LocalizedText("Employees"), new Loc
    }
}
```

— 课程1

- UaWebServer负责提供OPC UA Http服务
- 通过重载onStartUp对UA服务进行初始化
- 需将EmployeeTwinSpace注册到UaWebServer中，确保提供数字孪生服务
- 可通过设置UaServerConfigure对UA Server的服务参数进行设置
- 可通过resource中的application.properties中设置服务的参数，如URI，端口号，数据库连接等

```
@Service 0 个用法  @dingyan <dy_100000@163.com>
public class UaWebServer extends org.opcfoundation.webserver.service.UaWebServer {
    public UaWebServer() { super(); }

    @Override 0 个用法  @dingyan <dy_100000@163.com>
    public void onStartUp() {
        System.out.println("Start up server with NodeManager ");

        // Initialize OPC UA Web Server configuration
        configureServer();

        // Add opc ua data manager
        addNodeManager(new EmployeeDataManager());
    }

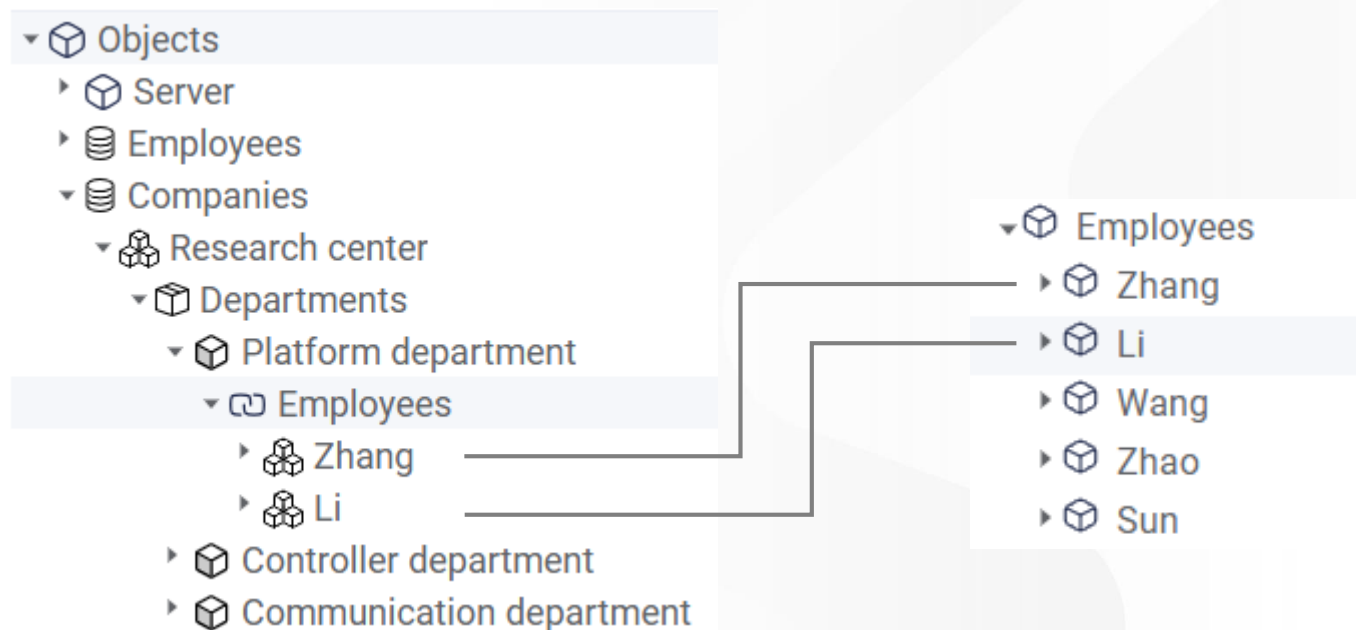
    @Override 0 个用法  @dingyan <dy_100000@163.com>
    public void onShutDown() { System.out.println("Shut down server with NodeManager"); }

    private void configureServer() 1 个用法  @dingyan <dy_100000@163.com>
    {
        UaServerConfigure configure = new UaServerConfigure();
        configure.setApplicationUri("EmployeeDemo");
        configure.setApplicationName(LocalizedText.english(text: "EmployeeDemo"));
        configure.setProductUri("EmployeeDemo");
        setServerConfigure(configure);
    }
}
```

— 课程1-总结

1. EmployeeDigitalTwinType: 负责体现员工的数字孪生
2. EmployeeRepositoryType: 负责提供员工的数字孪生列表
3. PersonalDataSubmodelType/EmployeeDataSubmodelType: 负责提供员工数字孪生的个人和企业信息
4. EmployeeDataSubmodelType::getCumulatedSalary: 通过getCumulatedSalary操作, 实现对员工累计工资的计算
5. SexEnumType: 通过定义性别枚举, 进一步说明性别字段数值的具体含义
6. EmployeeTwinSpace: 负责管理与员工和企业相关的数字孪生信息和服务
7. UA Web服务: 负责提供OPC UA WebService, 并将DigitalTwinSpace中的数据与服务提供给UA WebClient
8. Spring框架: OPC UA WebService基于Spring框架进行扩展, 因此遵循微服务架构原则, 并继承Spring框架的能力, 例如采用application.properties进行服务配置, 使用如@Server, @Component等注释

— 课程2



ID	Name	Comment	ParentDepartmentId
1	Research center	New tech R&D, including DCS/PLC/Industry soft...	NULL
2	Platform department	Software and platform development	1
3	Controller department	PLC programing tool and runtime development	1
4	Communication department	Industry protocol R&D	1

EmployeeId	DepartmentId
1	2
2	2
3	3
4	3
5	3

— 课程2

可视化

公司数字孪生定义

- ▼ DigitalTwinType
 - ▶ Employee
 - ▶ CompanyDigitalTwin

公司部门子模型定义

- ▼ SubmodelType
 - ▶ PersonalDataModel
 - ▶ EmployeeDataModel
 - ▶ DepartmentModel

公司数字孪生

- ▼ Companies
 - ▶ Research center

部门子模型

- ▼ Research center
 - ▼ Departments
 - ▶ Platform department
 - ▶ Controller department
 - ▶ Communication department

CompanyDigitalTwinType

- 用于表示公司数字孪生
- 继承自DigitalTwinType
- 在构造函数中添加 departmentSubmodel
- 重载onGetDescriptor, 返回数字孪生的名称和描述信息

```
public class CompanyDigitalTwinType extends DigitalTwinType { 4个用法
    public CompanyDigitalTwinType( 1个用法 @dingyan <dy_100000@163.co
        DepartmentSubmodelType departmentSubmodel,
        EmployeeTwinSpace twinSpace) {
        super( typelid: "CompanyDigitalTwinType", new LocalizedText("Com

        addSubmodel(
            departmentSubmodel,
            name: "Departments",
            new LocalizedText("Departments"),
            new LocalizedText("Company departments"));
    }

    public CompletableFuture<GetDescriptorResponse> onGetDescriptor(
        {...}
    }
}
```

DepartmentSubmodelType

- 负责提供公司的部门列表
- 继承自DynamicSubmodelType
- 重载onGetObjectElementList, 并调用数据库, 返回下级部门列表

```
/* Defining department submodel including list of highest level department
public class DepartmentSubmodelType extends DynamicSubmodelType { 5个用法 @dingyan

    public DepartmentSubmodelType(EmployeeTwinSpace twinSpace) 1个用法 @dingyan
    {
        super( typelid: "DepartmentSubmodelType", new LocalizedText("DepartmentModel"

    }

    // Return department general information such as name, description
    public CompletableFuture<GetObjectElementListResponse> onGetObjectElementList
    }
}
```


— 课程2

可视化

部门元素定义

- ▼  ElementType
 - ▶  ReferenceElementType
 - ▼  ElementCollectionType
 - ▶  Department

部门员工引用定义

- ▼  ElementType
 - ▼  ReferenceElementType
 -  DepartmentEmployee

部门与员工

- ▼  Platform department
 - ▼  Employees
 - ▶  Zhang
 - ▶  Li

DepartmentType

- 用于表示某一部门
- 继承自ElementCollectionType
- 重载onGetDescriptor, 返回数字孪生的名称和描述信息

```
public class DepartmentType extends ElementCollectionType { 4个用法
    public DepartmentType( 1个用法  @dingyan <dy_100000@163.com>
        DepartmentEmployeeReferenceType departmentEmployeeReferenceType,
        EmployeeTwinSpace twinSpace) {
        super( typeId: "DepartmentType", new LocalizedText("DepartmentType") );

        addReferenceElement(
            departmentEmployeeReferenceType,
            name: "Employees",
            new LocalizedText("Employees"),
            new LocalizedText("Employees Of department"),
            mandatory: true);
    }

    public CompletableFuture<GetDescriptorResponse> onGetDescriptor() {
        ...
    }
}
```

DepartmentEmployeeReferenceType

- 负责提供部门员工信息
- 继承自ReferenceElementType
- 重载onGetLinks, 并调用数据库, 返回部门员工列表

```
public class DepartmentEmployeeReferenceType extends ReferenceElementType
    public DepartmentEmployeeReferenceType(EmployeeTwinSpace twinSpace) {
        super( typeId: "DepartmentEmployeeReferenceType", new LocalizedText("DepartmentEmployeeReferenceType") );

        // Get employee of department
        @Override 0个用法  @dingyan <dy_100000@163.com>
        public CompletableFuture<GetLinkResponse> onGetLinks(GetLinkRequest request) {
            ...
        }
    }
}
```

— 课程2-总结

1. DepartmentType: 负责体现部门的信息，且需提供部门的员工信息
2. DepartmentEmployeeReferenceType: 负责提供部门员工的信息列表

— 课程3

▼ Servers

demo

▼ Objects

▼ Server

▼ Employees

▼ Zhang

▼ PersonalData

▼ EmployeeData

▼ GetCumulatedSalary

▼ Attendance

▼ SkillInfo

▼ Li

▼ Wang

▼ Zhao

▼ Sun

▼ Companies

▼ Types

▼ Views

Variables

References

Event History

历史记录

Attendance

开始时间 2026-01-01

结束时间 2026-01-10

搜索

Query

▼ BaseEventType

EventId

EventType

Time

Message

▼ EmployeeCheckInEventType

CheckIn

Location

Results (4)

Index	Time	Message
1	2026-01-05 15:53:00	N/A
2	2026-01-06 01:32:00	N/A
3	2026-01-06 16:05:32	Late
4	2026-01-07 01:32:00	N/A

	ID	Time	CheckIn	Location	Remark	EmployeeId
▶	1	2026-01-05 07:53:00	1	北京	NULL	1
	2	2026-01-05 17:32:00	0	北京	NULL	1
	3	2026-01-06 08:05:32	1	北京	迟到	1
	4	2026-01-06 17:32:00	0	北京	NULL	1
	5	2026-01-05 07:57:00	1	杭州	NULL	2
	6	2026-01-05 17:32:00	0	杭州	NULL	2
	7	2026-01-06 07:54:32	1	杭州	NULL	2
	8	2026-01-06 17:10:00	0	杭州	早退	2

— 课程3

可视化

考勤事件元素定义

- ElementType
 - ReferenceElementType
 - ElementCollectionType
 - ElementListType
 - EventElementType
 - EmployeeAttendanceElementType

打卡记录事件类型定义

- EventTypes
 - BaseEventType
 - EmployeeCheckInEventType

事件字段

Index	Name	Data Type	Value
1	CheckIn	Boolean	
2	Location	String	

EmployeeAttendanceElementType

- 用于表示模型能产生事件
- 继承自EventElementType
- 在构造函数中指定产生的事件类型 (EmployeeCheckInEventType)
- 重载onGetEvent，返回员工的打卡信息

```
public class EmployeeAttendanceElementType extends EventElementType {
    public EmployeeAttendanceElementType(EmployeeCheckInEventType ev) {
        super( typeId: "EmployeeAttendanceElementType", new LocalizedText("EmployeeAttendanceElementType") );
    }

    @Override
    public CompletableFuture<GetEventResponse> onGetEvent(GetEventRequest request) {
        // ...
    }
}
```

EmployeeCheckInEventType

- 负责定义事件的字段
- 继承自EventType
- 调用addField添加字段
- 编写generateEventData生成用于返回查询的事件数据

```
public class EmployeeCheckInEventType extends EventType {
    public static final String CheckIn = "CheckIn";

    public EmployeeCheckInEventType(DigitalTwinSpace space) {
        super( typeId: "EmployeeCheckInEventType", new LocalizedText("EmployeeCheckInEventType") );
        addField(CheckIn, new LocalizedText(CheckIn), new LocalizedText("EmployeeCheckInEventType"));
    }

    public static EventData generateEventData(EmployeeCheckIn record) {
        // ...
    }
}
```

— 课程2-总结

1. EmployeeAttendanceElementType: 负责表示对象可以进行事件查询
2. EmployeeCheckInEventType: 负责定义事件的具体字段

谢谢!

Intelligence For Excellence